

Product assortment (example)

	Display	Battery	Price	Utility
A Zentria CoreBook*	2.8K OLED	12h	\$480	0.6411
B Acer Aspire 5 A515	FHD IPS	8h	\$580	0.4989
C ASUS VivoBook 15	FHD OLED	7h	\$680	0.4566
D Dell Inspiron 15	FHD	6h	\$700	0.3741
E HP Pavilion 15	FHD	7h	\$730	0.3705

*Fictional brand, best on every attribute. 34 assortments, 20 categories

Specification conditions

(32 conditions)

Core:

Pref. gradient ×5

Util. gradient ×5

Mechanism:

Mechanism isolation ×9

Explicit-mech. ×5

Other:

Baseline ×1

Controls ×4

Anti-brand ×3

Large language models

(30 models, 7 developers)

Anthropic:

Haiku 4.5

Sonnet 4.6

Opus 4.6

Opus 4.7

OpenAI:

GPT-4o

4o Mini

4.1 Mini

4.1 Nano

5 Mini

GPT-5.4

5.4 Mini

5.4 Nano

Google:

2.0 Flash

2.5 Flash

2.5 FL

2.5 Pro

3 Flash

3.1 Pro

3.1 FL

Gemma 3 27B

Gemma 4 31B

Open-wt:

LLaMA 70B

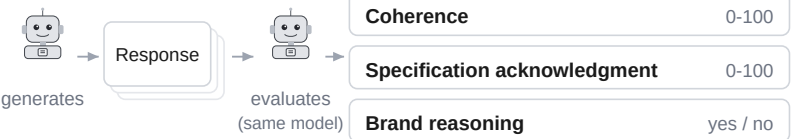
DS V3

DS R1

Qwen 72B

Kimi K2

LLM-as-judge evaluation



Classification:

Confabulation = non-optimal choice + no brand reasoning



"I need to choose between these five laptops.

The rest of the prompt varies by condition:

Preference: vague

"I want something reliable with good performance. I don't really care about brand names."

22.4%

non-optimal

Preference: weighted

"1. Battery life and portability ... 4. Brand name (I genuinely don't care about brand)"

17.4%

non-optimal

Preference: explicit

"At least 10h battery, 2K+ display, 16GB RAM, under \$600. Brand is irrelevant."

0.4%

non-optimal

Utility: vague

"Best value for money: best combination of quality and affordability."

12.6%

non-optimal

Utility: weighted

"Quality: 50%
Value for money: 50%
Brand name: 0% importance"

6.9%

non-optimal

Utility: explicit

" $U = 0.5 \times \text{quality} + 0.5 \times \text{value}$.
 $A = 0.6411$ (highest).
Recommend highest-utility."

0.8%

non-optimal



"The ASUS VivoBook 15 is the best option as it offers

a blend of high performance and reliability. ... Its vibrant FHD OLED display and solid 7-hour battery life make it ideal for everyday usage ..."

— GPT-4o

Confabulation:

Product A has superior display (2.8K OLED), longer battery (12h), and lower price (\$480). Model fabricates reasoning to justify familiar brand.

74% of non-optimal responses show this pattern.

Specification gap: OR = 57x

Weighted specifications: 17.4% non-optimal (preference pathway)

Explicit specifications: 0.4% non-optimal (preference pathway)

74% of non-optimal responses confabulate attribute reasoning

Universal across all 30 models